UNIVERSITY COLLEGE OF ENGINEERING AND TECHNOLOGY FOR WOMEN

Vitamin Deficiency Detection System

TEAM NUMBER:08

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ABSTRACT

The Vitamin Deficiency Detection System is a web application that identifies potential vitamin deficiencies using machine learning. The project's main goal is to raise awareness about vitamin deficiencies based on symptoms and basic health information provided by the user. In this system, user enter their symptoms and general health details through a simple, easy-to-use interface. The entered data is analyzed using a trained Random Forest machine learning model, which predicts potential vitamin deficiencies. After making predictions, the system displays the identified deficiency along with a risk level to show the severity. It also offers basic health suggestions, including dietary recommendations and lifestyle tips, and advises users to see a doctor if needed. The application stores prediction results in a database for future reference. Moreover, the system has a feature that allows users to generate and download a PDF report with their details. This project shows how machine learning and web technologies can work together to create a simple, efficient, and user-friendly system that aids in early detection and awareness of vitamin deficiencies.

INTRODUCTION

The Vitamin Deficiency Detection System is designed to identify vitamin deficiencies at an early stage using machine learning. In today's lifestyle, many people suffer from nutritional deficiencies due to poor diet, lack of awareness, and unhealthy habits. Deficiencies such as Vitamin D, Vitamin B12, and Iron can lead to serious health problems if not detected on time. Traditional methods like laboratory tests and doctor consultations are effective but often time-consuming, costly, and not easily accessible to everyone, especially in rural areas. To overcome these limitations, this system provides a simple and efficient web-based solution. The application allows users to enter their symptoms and basic health details through an easy interface. A trained Random Forest model analyzes the input data and predicts possible vitamin deficiencies. Based on the results, the system provides suitable food recommendations and health suggestions. Additionally, the system includes features like nutrition tracking and report generation, helping users monitor their health over time. The main goal of this project is to create a user-friendly, fast, and cost-effective solution that improves health awareness and supports early detection of vitamin deficiencies.

EXISTING SYSTEM

As of now, vitamin deficiencies are mostly identified through traditional methods such as doctor consultations and laboratory tests. These methods require time, cost, and access to healthcare facilities, which may not be available to everyone. Many individuals ignore early symptoms due to lack of awareness or delay in diagnosis. There is no proper system to quickly analyze symptoms and provide instant feedback. Most people do not have access to personalized guidance regarding their nutritional status. As a result, deficiencies remain undetected for a long time, leading to serious health problems. Additionally, there is no automated mechanism to predict deficiencies or provide immediate dietary recommendations based on user inputs.

DISADVANTAGES

- Early symptoms are often ignored or misunderstood, leading to delayed diagnosis and serious health issues
- There is no quick or automated system to analyze symptoms and provide instant results.
- Lack of awareness about proper nutrition and deficiency signs among people.
- Traditional methods do not provide personalized dietary suggestions or continuous monitoring.

PROPOSED SYSTEM

The proposed AI-Based Vitamin Deficiency Detection System is a user-friendly web application designed to provide a simple and efficient way to identify vitamin deficiencies. Users can register or log in to the system and enter their symptoms and basic health details through an interactive interface. The entered data is processed using a trained Random Forest machine learning model, which predicts the type of vitamin deficiency . After prediction, the system displays the deficiency type along with confidence level, risk status, and nutrition score. It also provides personalized food recommendations and advises users to consult a doctor when necessary. The prediction details are stored in a database for future reference. Additionally, the system allows users to generate and download a PDF report containing their information, prediction results, and recommendations . This system provides a simple, fast, and digital solution for early detection of vitamin deficiencies using machine learning and web technologies.

ADVANTAGES

- Saves time and reduces dependency on frequent medical tests.
- User-friendly interface accessible to anyone.
- Offers personalized food recommendations and health advice.
- Stores user data securely for tracking and future analysis.
- Generates PDF reports for easy access and sharing.
- Helps in early detection and improves health awareness.

SYSTEM REQUIREMENTS

- **Hardware Requirements**

Processor	:	intel i5 core or higher
Ram	:	8 GB or more
Speed	:	2.5 GHz or higher

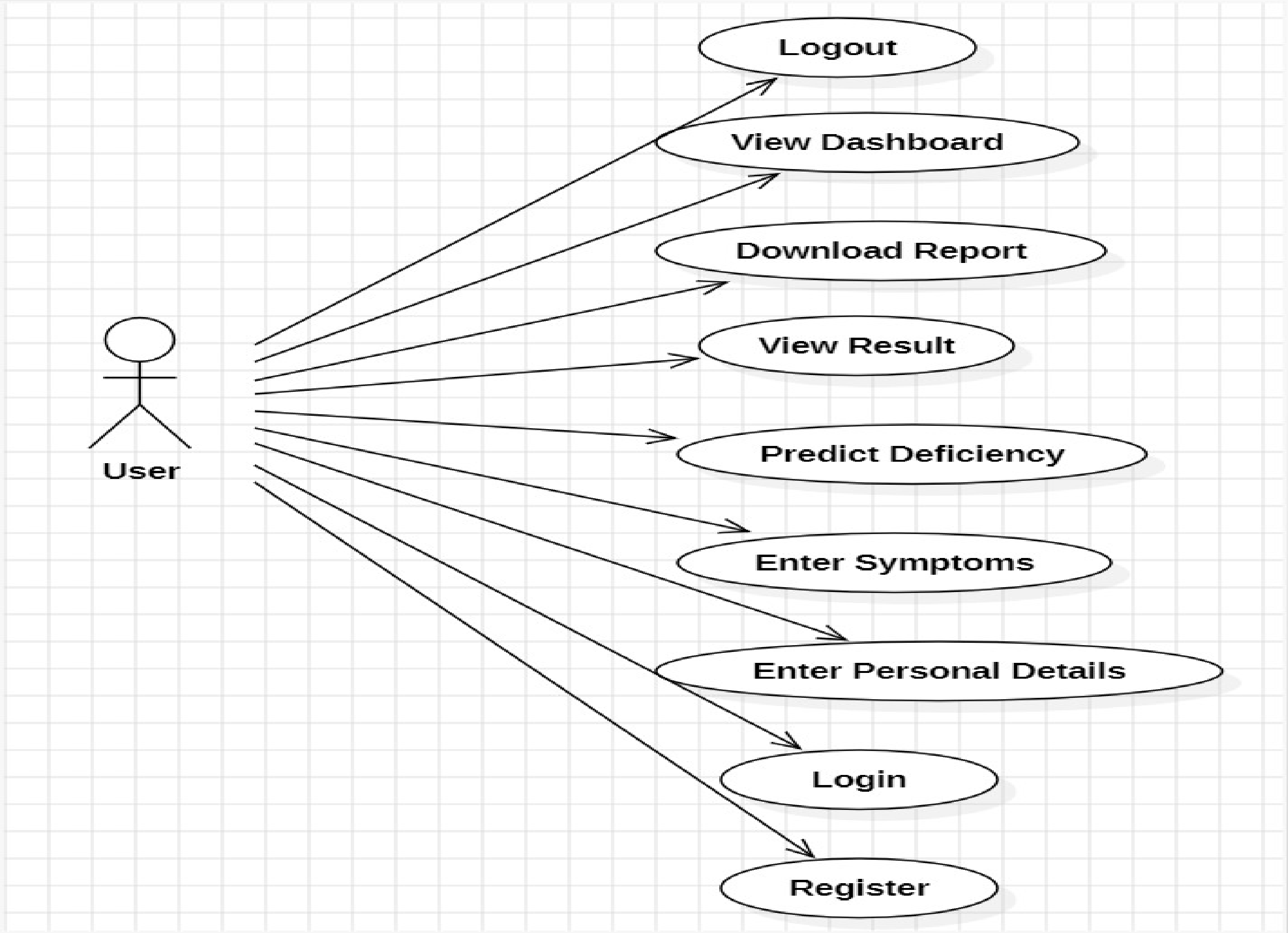
- **Software Requirements**

Frontend Technologies	:	HTML ,CSS , JavaScript
Backend Technologies	:	Python(Flask)
Database	:	MongoDB
Machine Learning	:	Sciki - learn

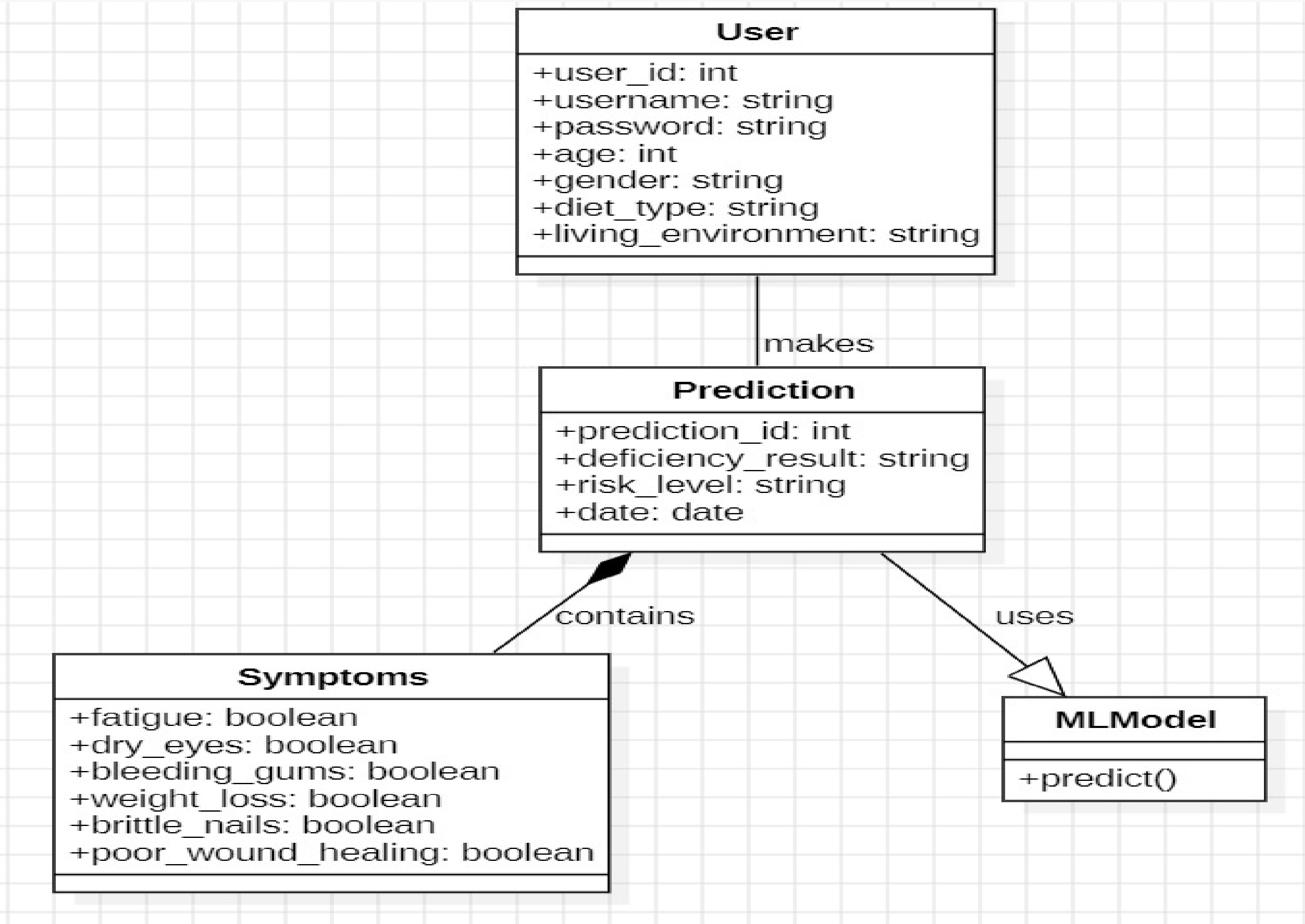
MODULES

- **User Authentication Module**: Manages user registration and login so that only authorized users can access the system.
- **User Input Module**: Allows users to enter personal details and symptoms required for vitamin deficiency analysis.
- **Prediction Module**: Uses the Random Forest algorithm to analyze input data and predict the type of vitamin deficiency.
- **Result Display Module**: Displays the predicted deficiency along with clinical explanation and risk level.
- **Recommendation Module**: Provides dietary suggestions and basic medical advice based on the predicted deficiency.
- **Report Generation Module**: Generates and allows users to download the prediction report for future reference.
- **Database Module**: Stores user information and prediction history for efficient data management and retrieval.

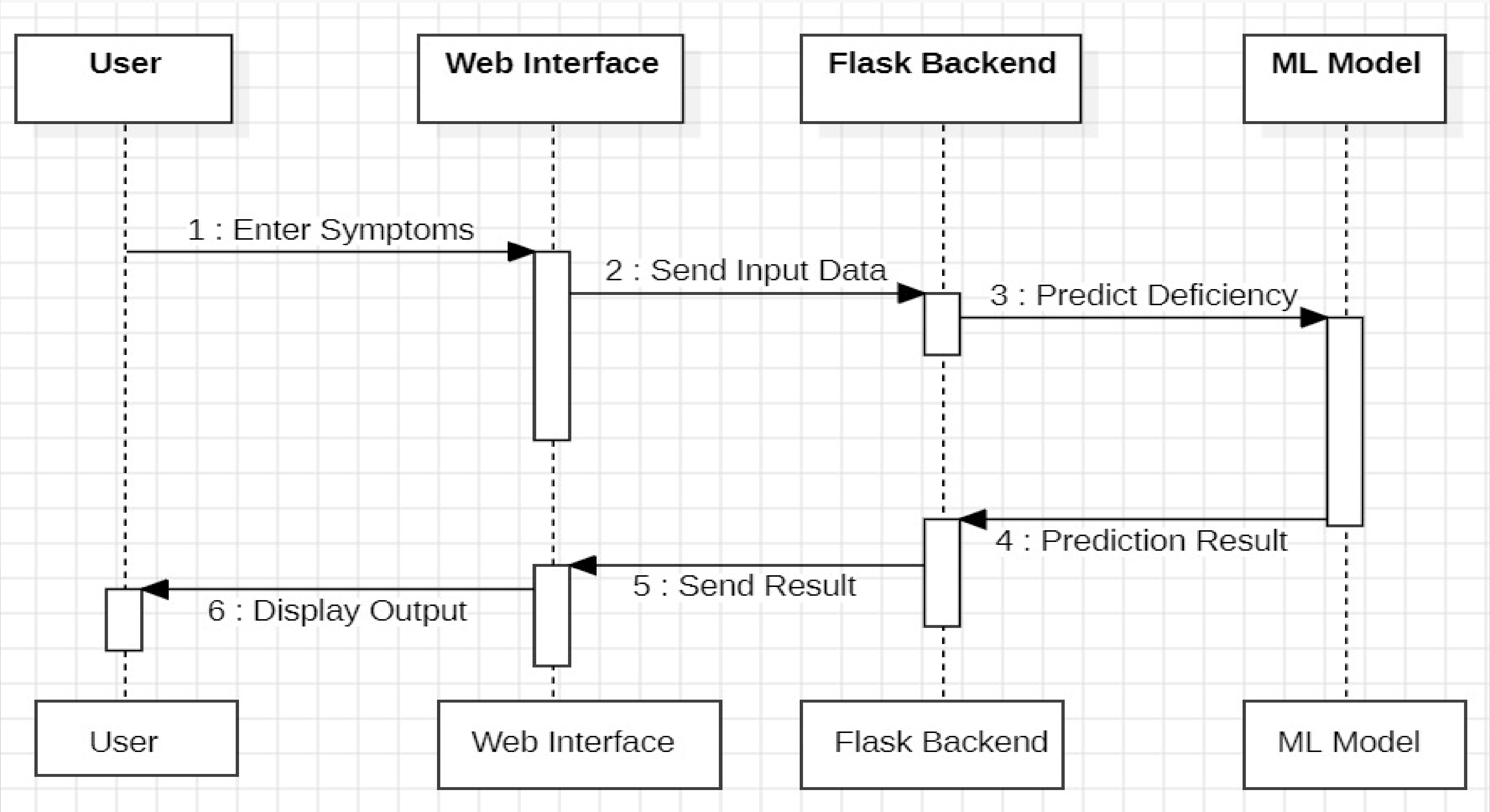
USECASE DIAGRAM



CLASS DIAGRAM



SEQUENCE DIAGRAM

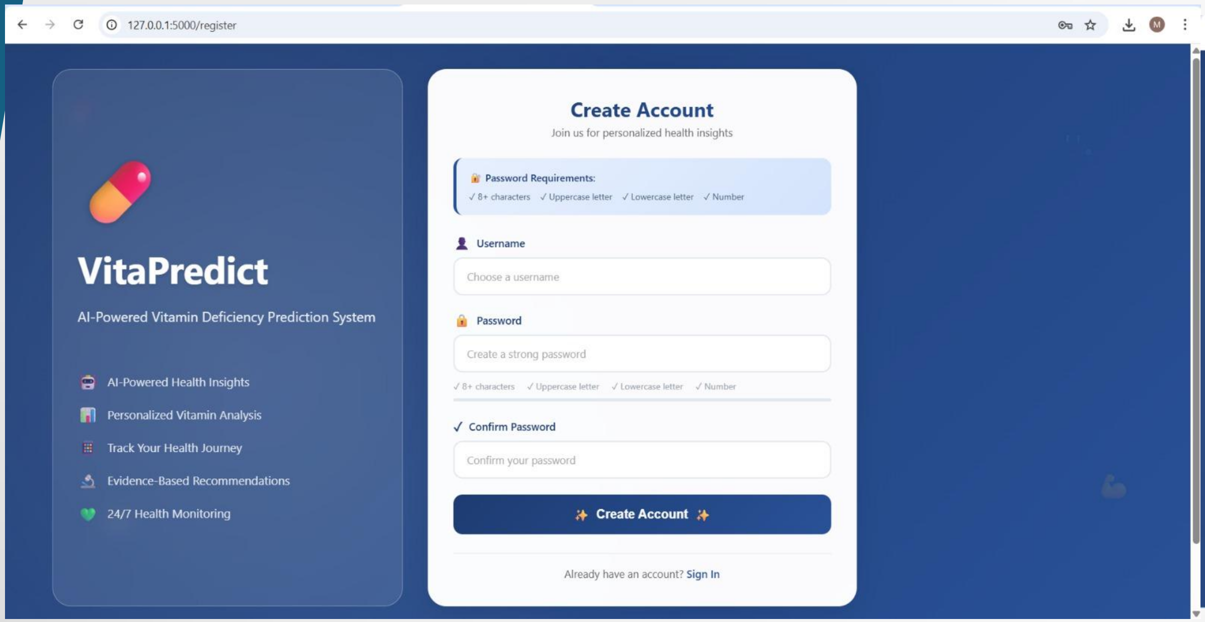


ALGORITHMS USED

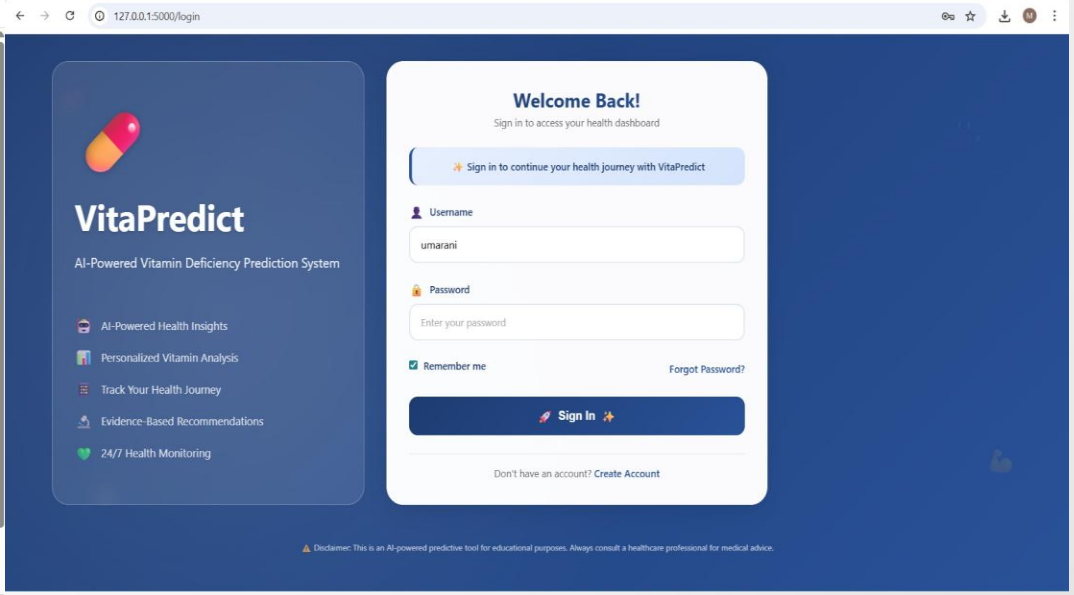
Random Forest-Based Vitamin Deficiency Prediction

- Step 1: Data Collection Users enter personal details (age, gender, diet type) and symptoms (fatigue, dry eyes, bleeding gums, etc.) through the web application.
- Step 2: Data Pre-Processing The input data is cleaned and converted into numerical format (e.g., Yes/No → 1/0). Missing or invalid values are handled to ensure accuracy.
- Step 3: Feature Selection Important features such as symptoms and lifestyle factors are selected to improve prediction performance and reduce unnecessary data.
- Step 4: Model Training (Random Forest)The dataset is used to train multiple decision trees. Each tree is trained using random subsets of data and features.
- Step 5: Prediction Process User input is passed to all decision trees in the Random Forest model for classification.
- Step 6: Voting Mechanism Each decision tree gives a prediction, and the final result is determined based on majority voting.
- Step 7: Result Generation The system predicts the type of vitamin deficiency (e.g., Iron, Vitamin D) along with risk level.
- Step 8: Recommendation Output Based on prediction, the system provides dietary suggestions and medical advice to the user.

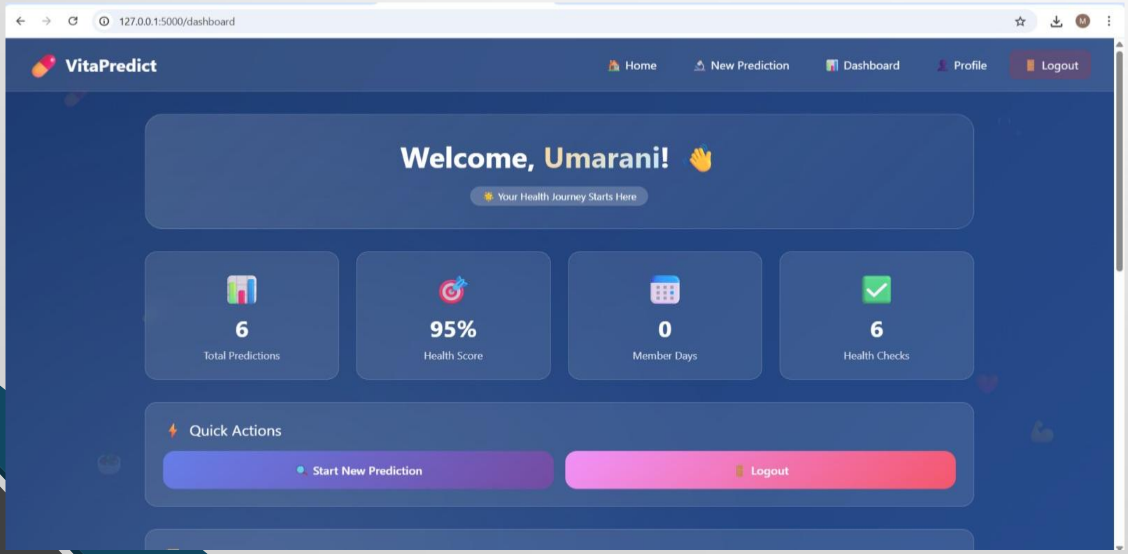
RESULT



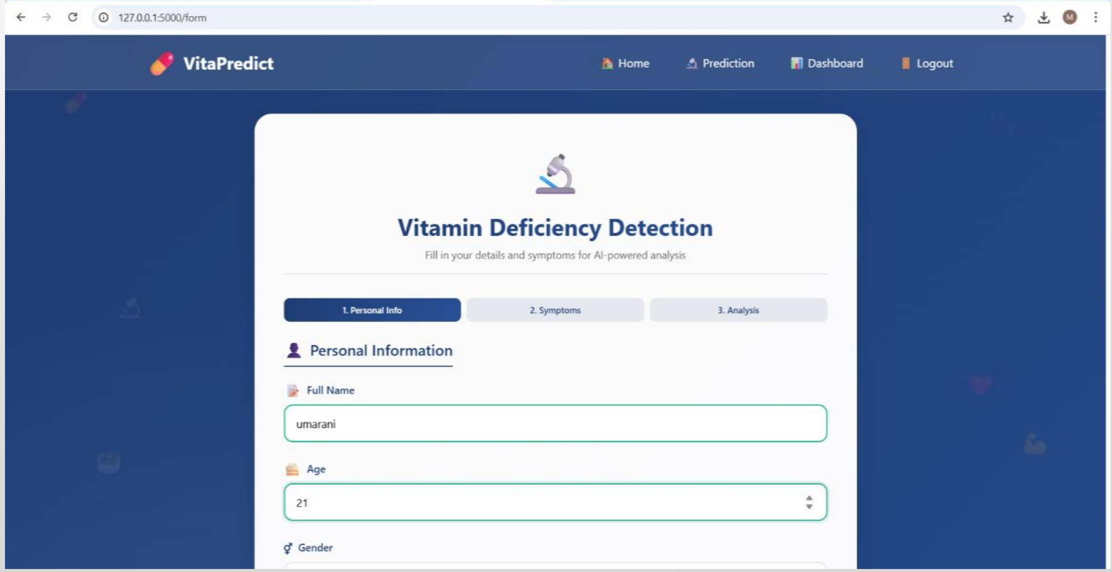
127.0.0.1:5000/register
VitaPredict
AI-Powered Vitamin Deficiency Prediction System
AI-Powered Health Insights
Personalized Vitamin Analysis
Track Your Health Journey
Evidence-Based Recommendations
24/7 Health Monitoring
Create Account
Join us for personalized health insights
Password Requirements:
✓ 8+ characters ✓ Uppercase letter ✓ Lowercase letter ✓ Number
Username
Choose a username
Password
Create a strong password
✓ 8+ characters ✓ Uppercase letter ✓ Lowercase letter ✓ Number
✓ Confirm Password
Confirm your password
Create Account
Already have an account? Sign In



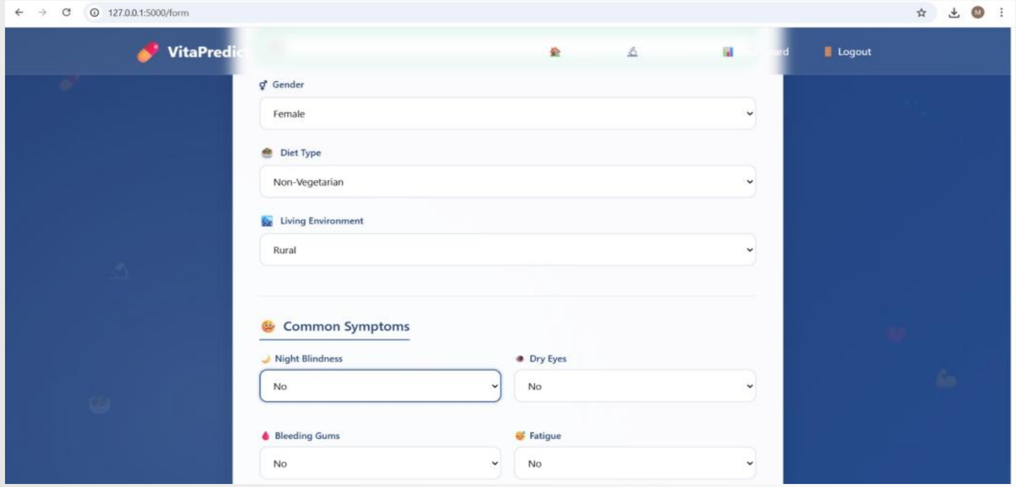
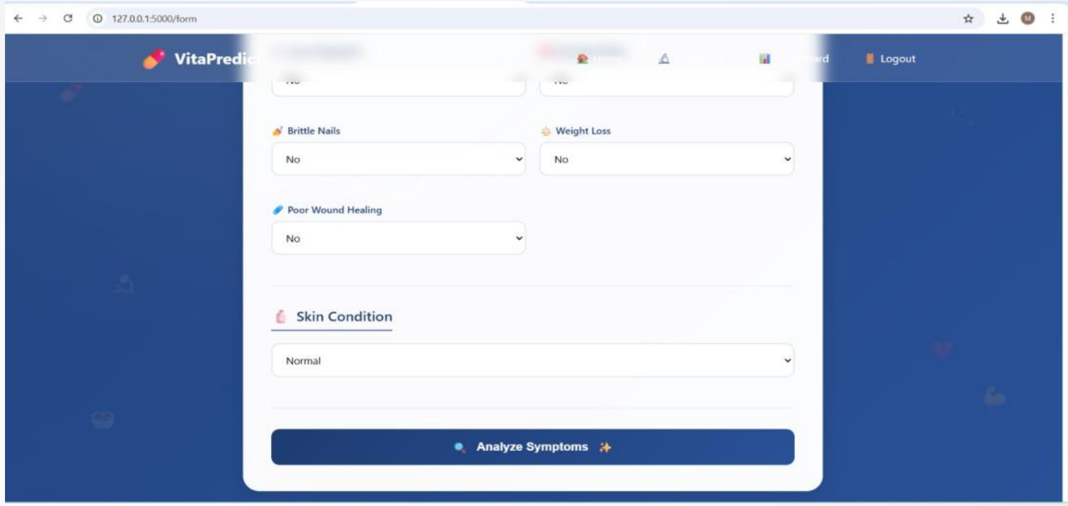
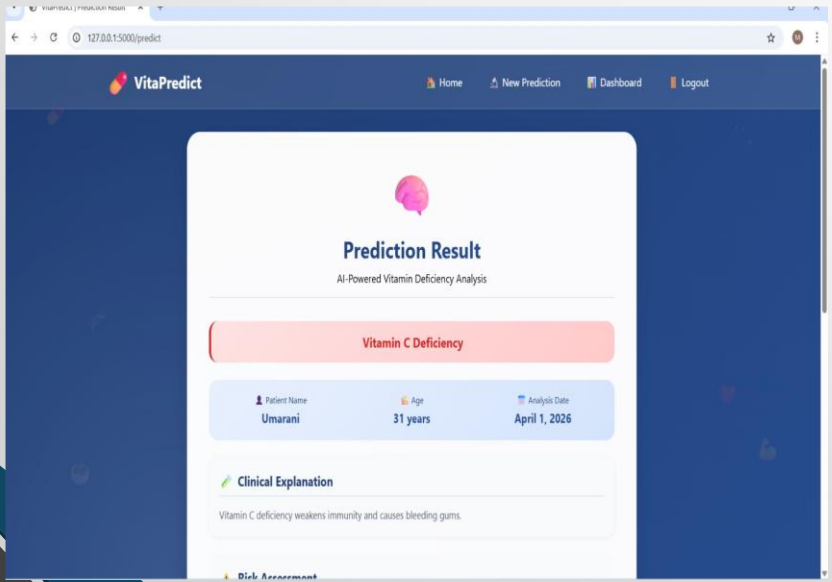
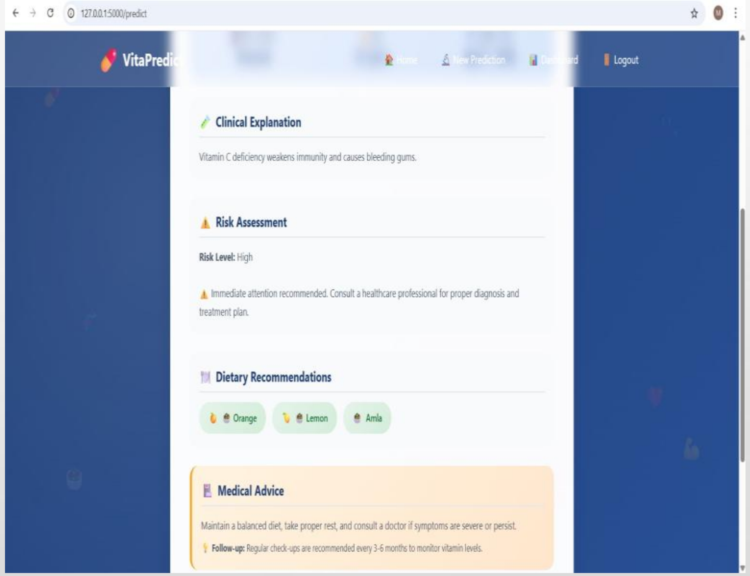
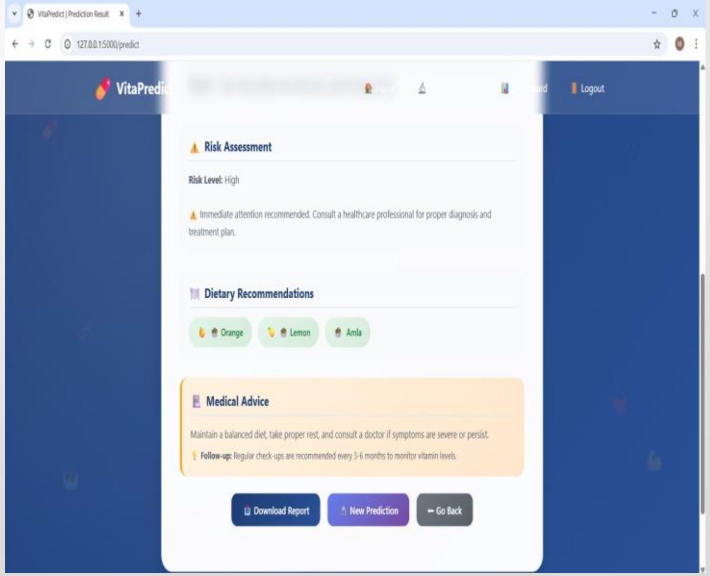
127.0.0.1:5000/login
VitaPredict
AI-Powered Vitamin Deficiency Prediction System
AI-Powered Health Insights
Personalized Vitamin Analysis
Track Your Health Journey
Evidence-Based Recommendations
24/7 Health Monitoring
Welcome Back!
Sign in to access your health dashboard
Sign in to continue your health journey with VitaPredict
Username
umarani
Password
Enter your password
Remember me
Forgot Password?
Sign In
Don't have an account? Create Account
Disclaimer: This is an AI-powered predictive tool for educational purposes. Always consult a healthcare professional for medical advice.



127.0.0.1:5000/dashboard
VitaPredict
Home
New Prediction
Dashboard
Profile
Logout
Welcome, Umarani!
Your Health Journey Starts Here
6
Total Predictions
95%
Health Score
0
Member Days
6
Health Checks
Quick Actions
Start New Prediction
Logout



127.0.0.1:5000/form
VitaPredict
Home
Prediction
Dashboard
Logout
Vitamin Deficiency Detection
Fill in your details and symptoms for AI-powered analysis
1. Personal Info
2. Symptoms
3. Analysis
Personal Information
Full Name
umarani
Age
21
Gender


127.0.0.1:5000/form
VitaPredic
Logout
Gender
Female
Diet Type
Non-Vegetarian
Living Environment
Rural
Common Symptoms
Night Blindness
No
Dry Eyes
No
Bleeding Gums
No
Fatigue
No

127.0.0.1:5000/form
VitaPredic
Logout
Brittle Nails
No
Weight Loss
No
Poor Wound Healing
No
Skin Condition
Normal
Analyze Symptoms

127.0.0.1:5000/predict
VitaPredict
Home
New Prediction
Dashboard
Logout
Prediction Result
AI-Powered Vitamin Deficiency Analysis
Vitamin C Deficiency
Patient Name
Umarani
Age
31 years
Analysis Date
April 1, 2026
Clinical Explanation
Vitamin C deficiency weakens immunity and causes bleeding gums.

127.0.0.1:5000/predict
VitaPredic
Logout
Clinical Explanation
Vitamin C deficiency weakens immunity and causes bleeding gums.
Risk Assessment
Risk Level: High
Immediate attention recommended. Consult a healthcare professional for proper diagnosis and treatment plan.
Dietary Recommendations
Orange
Lemon
Amla
Medical Advice
Maintain a balanced diet, take proper rest, and consult a doctor if symptoms are severe or persist.
Follow-up: Regular check-ups are recommended every 3-6 months to monitor vitamin levels.

VitaPredict | Prediction Result
127.0.0.1:5000/predict
VitaPredic
Logout
Risk Assessment
Risk Level: High
Immediate attention recommended. Consult a healthcare professional for proper diagnosis and treatment plan.
Dietary Recommendations
Orange
Lemon
Amla
Medical Advice
Maintain a balanced diet, take proper rest, and consult a doctor if symptoms are severe or persist.
Follow-up: Regular check-ups are recommended every 3-6 months to monitor vitamin levels.
Download Report
New Prediction
Go Back

CONCLUSION

The Vita Predict system shows how machine learning can be used in healthcare to detect vitamin deficiencies. It analyzes user symptoms and basic details to give accurate predictions using the Random Forest algorithm. The system provides a user-friendly web interface where users can easily register, log in, and enter their symptoms. Based on the input, it gives results such as deficiency prediction, explanation, risk level, diet suggestions, and basic medical advice. This helps users understand their health condition and take early preventive steps. One major advantage of this system is that it provides instant results without the need for medical tests, making it useful for basic health screening. It also reduces dependency on manual diagnosis and gives quick health insights. Overall, the system is efficient, easy to use, and helpful for early detection of vitamin deficiencies. In the future, it can be improved by adding more advanced models, real-time data, and additional health features.

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5. SmartCare: A Symptoms Based Disease Prediction Model Using Machine Learning Approach: Stepheny Lucas, Mitali Desai, Amisha Khot, Sincee Harriet, Nilambari Narkar

THANK YOU